The sensor array is laid out so that the pressure sensors measure the air drop directly inside the air pathway.

They use something called a distributed hierarchical architecture:

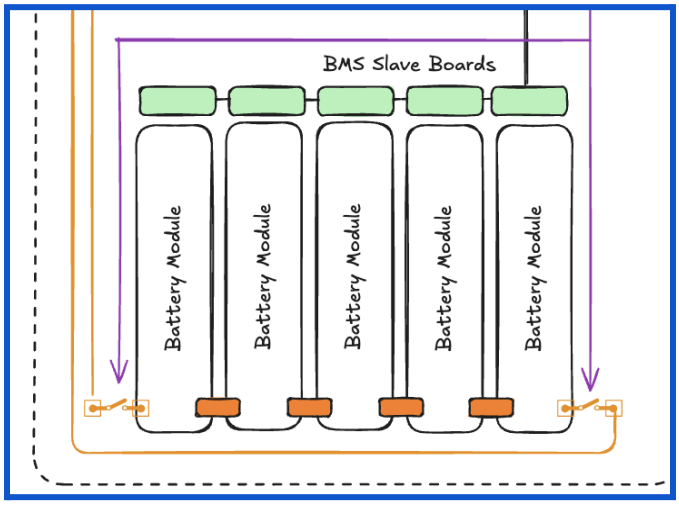
- CAN Bus:
    - Pressure and temperature values are broadcast onto CAN
    - The safety data is assigned the *lowest* identifier so that it can override general data packets
- One-Wire Digital Cascading:
    - Sensors are wired on a single master bus. Every sensor has a 64-bit code → central controller can "talk-to" dozens of parallel nodes on the singular copper track.
        - "Restricted to specific isolated topologies" (i have no idea what that means)
- Daisy Chain Bus:
    - Analog Front-End chips are linked together in a loop instead of running separate wire pairs from every single module to the main microcontroller.
    - Data travels linearly
    - Comm protocol is differential pulse signals over a single twisted pair of wires
        - What are AFE chips?

Options for Sensor Array:

- Bosch BMP 384 or BMP390: Absolute Pressure Sensors
    - Barometric sensors that also provide ambient temperature readings!
        - Problem: the BMP390 only has 2 I2C adresses
    - It supports high-speed SPI, so we can switch the architecture to SPI.
    - Supports the 3 standard wire lines: SCK, MISO and MOSI

- Sensirion SDP810-500 Pa or Würth WSEN-PDMS: Differential Pressure Sensors
    - High-precision differential pressure sensors that include on-chip temperature sensors.
    - Support 16-bit digital outputs over I2C or SPI protocols (SPI is preferred

Communication Network Architecture:



I noticed that the team is using the **BMS Slave Interface** approach (from the diagram ya'll showed in the workshops)

- Each battery module has its own dedicated Battery Front-End monitoring chip
- Off the shelf battery management system → bypass microcontrollers completely